

Lab 03.1: General performance of a motor vehicle

17/10/2025

Abstract

This report outlines a methodology for evaluating the general performance of a motor vehicle under different loading conditions. The procedure is structured as follows:

- **Resistance Modeling:** determination of resistance forces (aerodynamic, rolling, and slope) and estimation of transmission component efficiencies
- **Gear Ratio Definition:** calculation of the highest gear ratio to achieve maximum speed and determination of the lowest ratio to ensure climbing capacity on critical slopes through the tangency of power curves
- **Ratio Synchronization:** calculation of intermediate gears based on a geometric progression and comparison with manufacturer data
- **Longitudinal Dynamics Analysis:** evaluation of acceleration time within a defined speed range using a discrete-step numerical approach to integrate non-linear equilibrium equations

This approach allows for the quantification of how load variations impact the vehicle's top speed and acceleration performance.

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1 Introduction

The aim of the assignment is to assess the general performance of an average production car. More specifically the car selected for this purpose is the **FIAT Panda of 2010**. The car data are found on specific websites¹.



Figure 1

1.1 General performance meaning

The definition of “general performance” of a car means:

1. Maximum speed of the vehicle
2. Acceleration

1.2 Problem description

To compute the maximum speed and the acceleration of the vehicle in different operating conditions the following approach is presented. The different conditions cited above are different slopes and operating loads of the car.

Assumptions

- Vehicle presents as point, the mass is concentrated
- All forces are applied in the concentrated point
- The speed is constant

¹ <https://zeperfs.com/en/fiche12496-fiat-panda-iii-1-2.htm> for the torque curve
https://www.media.stellantis.com/uploads/em/2012/FIAT/FILES/121016_F_panda_techspecification.pdf for the technical sheet
<https://www.ultimatespecs.com/car-comparator/40432/Fiat-Panda-2012.html> for other technical data

Model

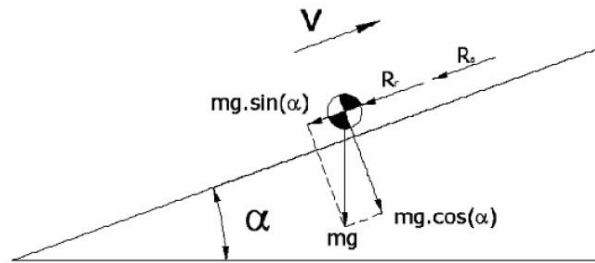


Figure 2

- m : vehicle mass
- v : constant speed
- α : ground inclination

Forces and power

- Resistance due to slope

$$R_i = P_t = P \sin \alpha$$

- Rolling resistance

$$R_0 = mg \cos(\alpha) (f_0 + f_2 * v^2)$$

- f_0 and f_2 consider the effect of the road surface
 - For f_0 a table is given
 - $f_2 = 0.5 \cdot 3.6^2 \times 10^{-6}$

- Aerodynamic resistance

$$R_a = 0.5 \rho C_x S v^2$$

- Resultant force

$$R_R = R_i + R_0 + R_a$$

- Power required

$$W_R = R_R v$$

- Available power

$$W_e = W_d \eta_G \eta_F \eta_D$$

- η_{G_i} : gear box efficiency of the i -th gear
- η_P : final drive efficiency
- η_D : transaxle efficiency (for double axle traction vehicle)

2 Assignment resolution

2.1 Passenger car choice

As presented in the introduction, the FIAT Panda of 2010 is chosen for this assignment.

2.2 Identification of the main features

Main features of the FIAT Panda:

- Mass
 - Running order: 1023.9 kg
 - Full load: 1308.3 kg
- Wheelbase: 2.3 m
- Maximum cross-sectional area and drag coefficient: $S * C_x = 0.69\text{ m}^2$
- Tire characteristics
 - Rolling radius: 0.292 m
 - Radial stiffness: $200\,000\text{ N/m}$
- Engine characteristics
 - $C_m = C_m(n)$

2.3 Estimation of transmission ratios and efficiencies

From the manufacturer, the following data are presented:

- Transmission ratios: $[3.909\ 2.158\ 1.480\ 1.121\ 0.897]$
- Efficiencies
 - Gearbox: 0.96, corresponding to helical gears
 - Final drive efficiency: 0.93, corresponding to straight bevel gears

2.4 Resistance power and engine power

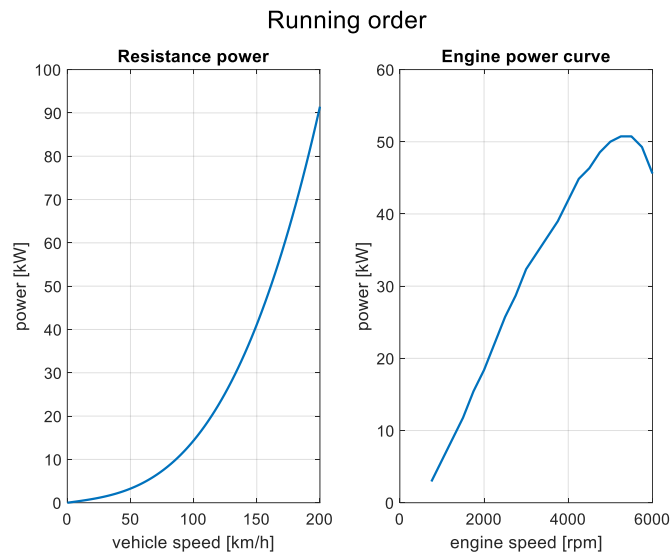


Figure 3

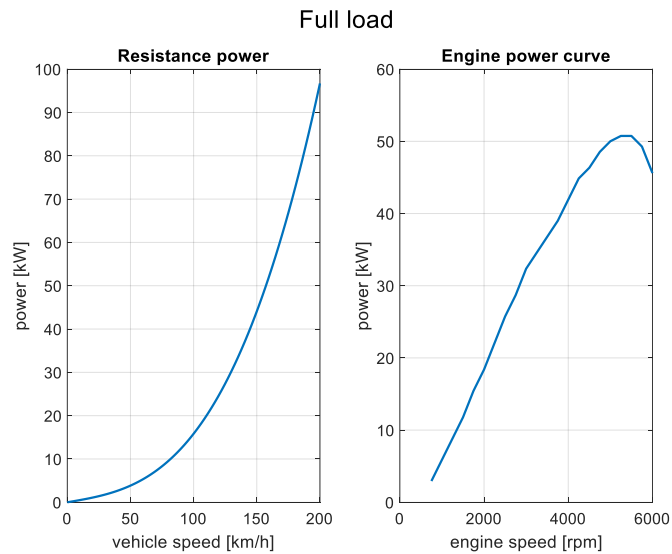


Figure 4

For both running order and full load conditions resistance power and engine power curve are computed.

- Resistance power
 - Higher in for the case of the fully loaded car
- Engine power torque
 - Same for both cases

- It presents a peak for about 5300rpm, value used later to compute the maximum speed

2.5 Ratio of the highest gear

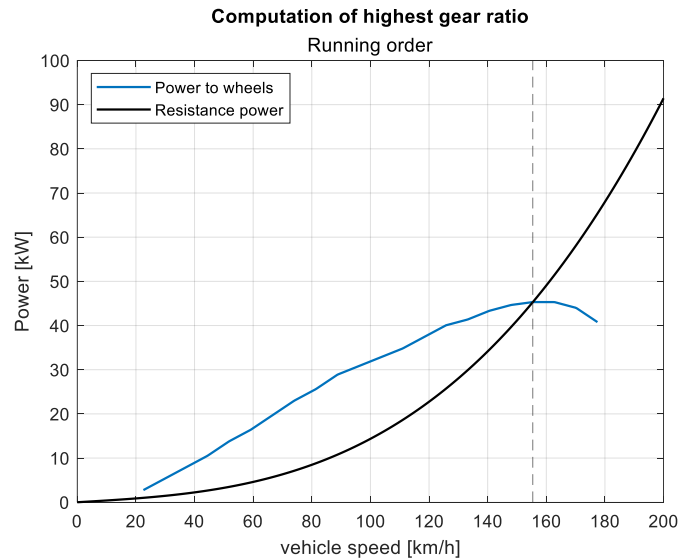


Figure 5

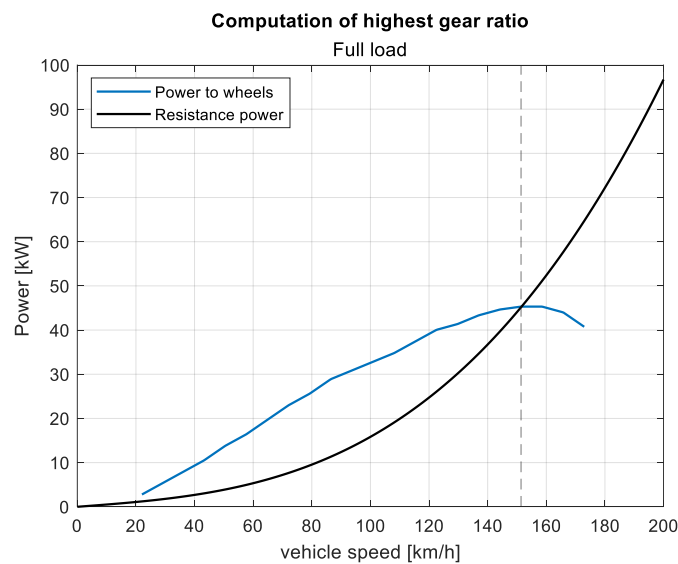


Figure 6

- The computation of the highest gear ratio is done by means of the maximum speed. To obtain the maximum achievable speed, the car must exploit its maximum power at the speed that can balance it out. So, there is a unique gear ratio that allows the car to travel at that speed, with the RPM corresponding to the maximum engine power.

- Maximum speed
 - Running order: 155.39 *km/h*
 - Full load: 151.38 *km/h*
- Higher gear ratio:
 - Running order: 1.0353
 - Full load: 1.05
- Different cases: the car with the higher load presents a slightly lower maximum speed, the difference is low as maximum speed is mainly an aerodynamic concern.

2.6 Ratio of the lowest gear

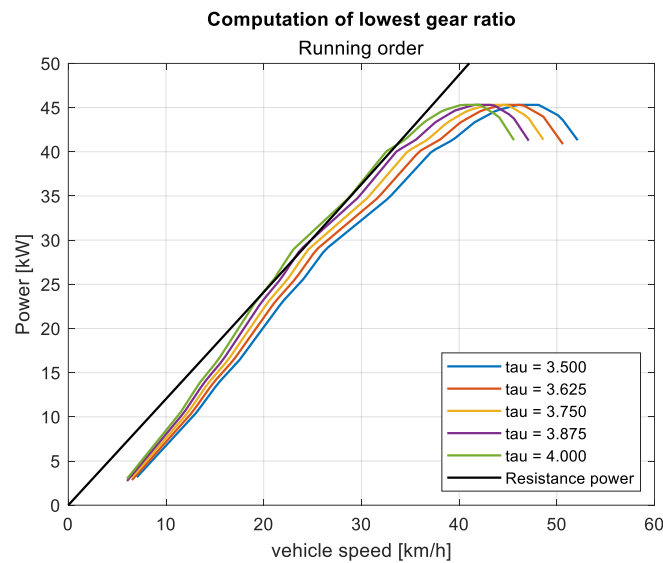


Figure 7

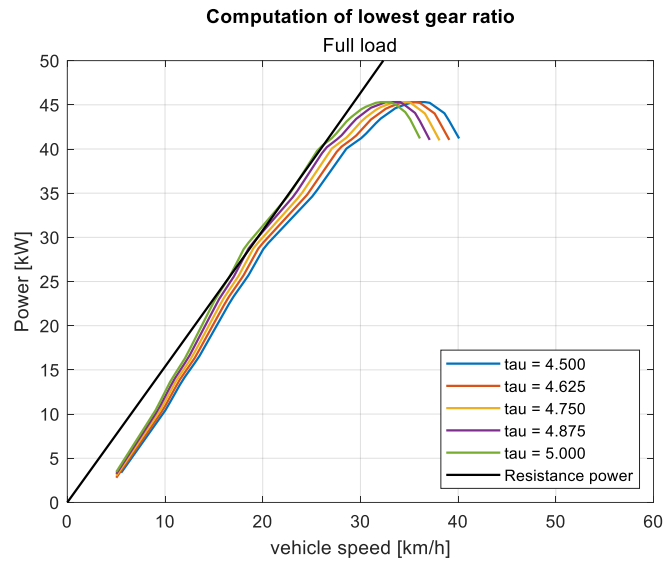


Figure 8

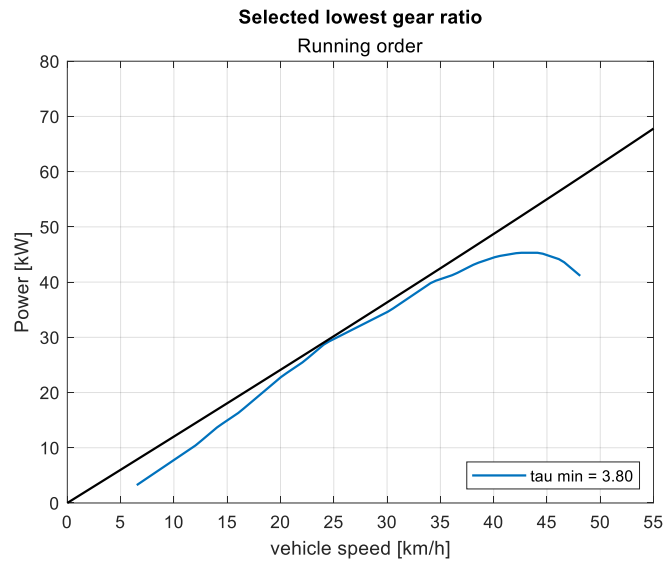


Figure 9

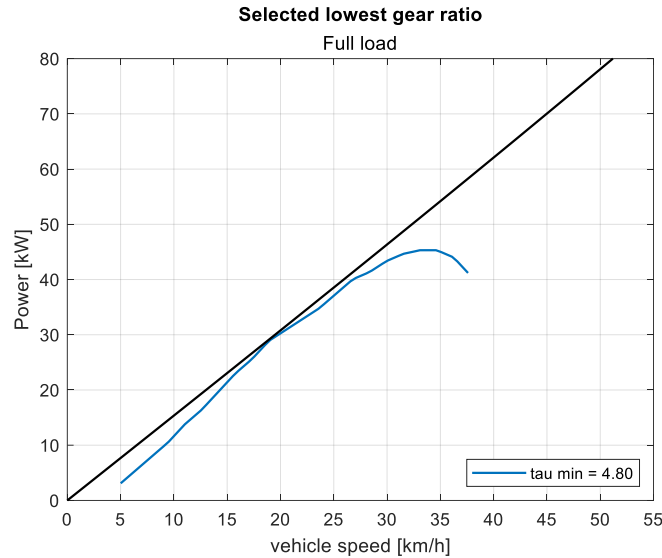


Figure 10

The requirements for the lower gear ratio given in the assignment are that:

- The car must be able to climb a maximum slope of 45%, which computationally corresponds to 46% to guarantee correct operational behavior of the vehicle
- The vehicle speed is about 7-12 km/h at 800 RPM

As presented in Figure 7 and Figure 8, the computation of the lowest gear ratio is done graphically. The correct gear ratio is the one for which the power output curve is tangent to the power required curve.

Lower gear ratios values:

- Running order: 3.8
- Full load: 4.8

2.7 Intermediate gears ratio

The intermediate gears are roughly equally spaced according to geometric progression. This ensures angular speed variation remains constant while passing from one gear ratio to the other.

The following formulas is used in this purpose:

$$\tau = \frac{\tau_{min}}{\tau_{max}} = cost$$

$$\tau_t^{\frac{1}{n-1}} = \frac{\tau_{i-1}}{\tau_i}$$

- Where:

- n : number of gear ratios
- τ_t : ratio of minimum and maximum gear ratios

Comparison with the manufacturer data

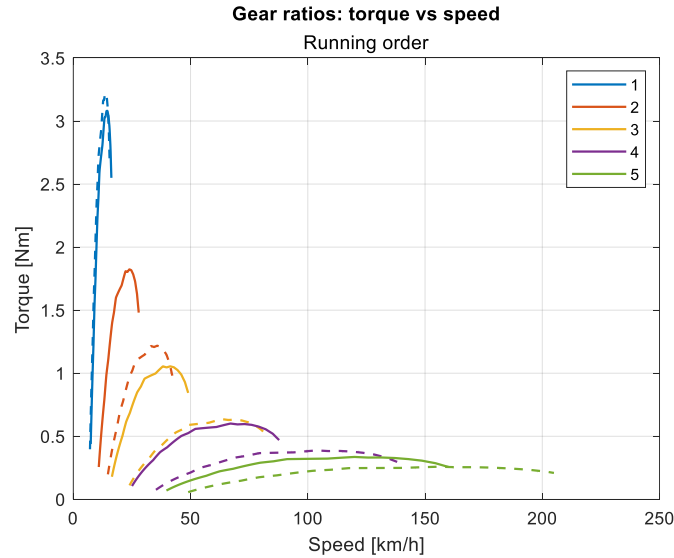


Figure 11

In this figure the ratios evaluated with the formulae are confronted with the ones given by the constructor. While the top gear matches almost perfectly, all the other ones are slightly shifted and the mismatch with the first gear has a relevant effect in the evaluations of the other ones.

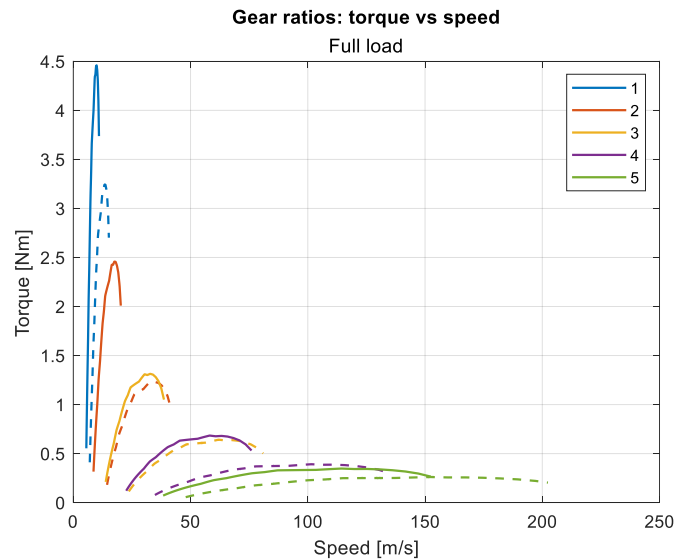


Figure 12

In Figure 12 instead, while the maps given by the manufacturer remain unchanged, the ones evaluated at full load show a higher request for torque at smaller speeds. This is due to the higher resistance forces that act on the car itself.

Gear	Manufacturer data	Computed ratios (Running Order)	Computed ratios (Full Load)
1	4.1	3.8	4.8
2	2.158	2.7454	3.2822
3	1.480	1.9835	2.2444
4	1.121	1.4330	1.5347
5	0.897	1.0353	1.0495

Here the evaluated ratios are reported. At full load they are higher once again because the resistance forces that need to be counterbalanced by the engine are higher.

2.8 Acceleration from 70 km/h to 120 km/h

The computation of the total acceleration time can be difficult analytically, since the contribution of the equilibrium equation are nonlinear.

Equilibrium:

$$m\ddot{a} = F_{drive} - F_{res}$$

- Both F_{drive} and F_{res} are nonlinear

To compute the total acceleration time from 70km/h to 120km/h, small steps of increasing speed are considered. Inside each step the acceleration is considered constant and obtained through the equilibrium presented above. Then, for each step, the delta time is computed and summed up to obtain the total time.

The plots presented help represent the computation.

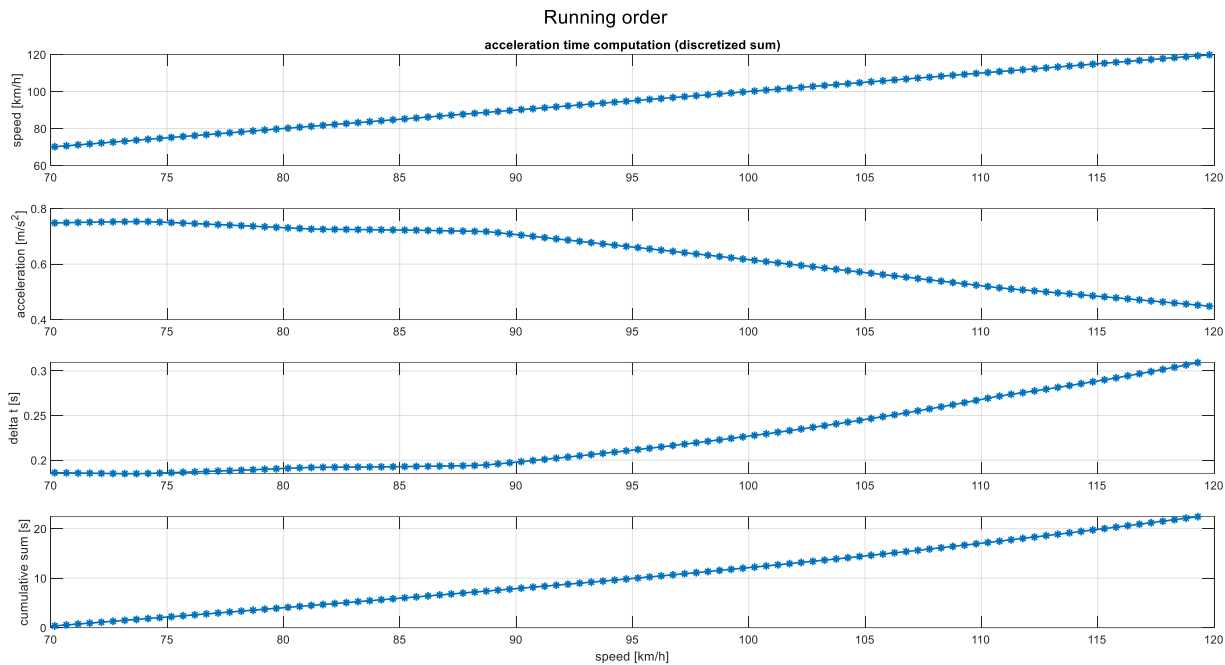


Figure 13

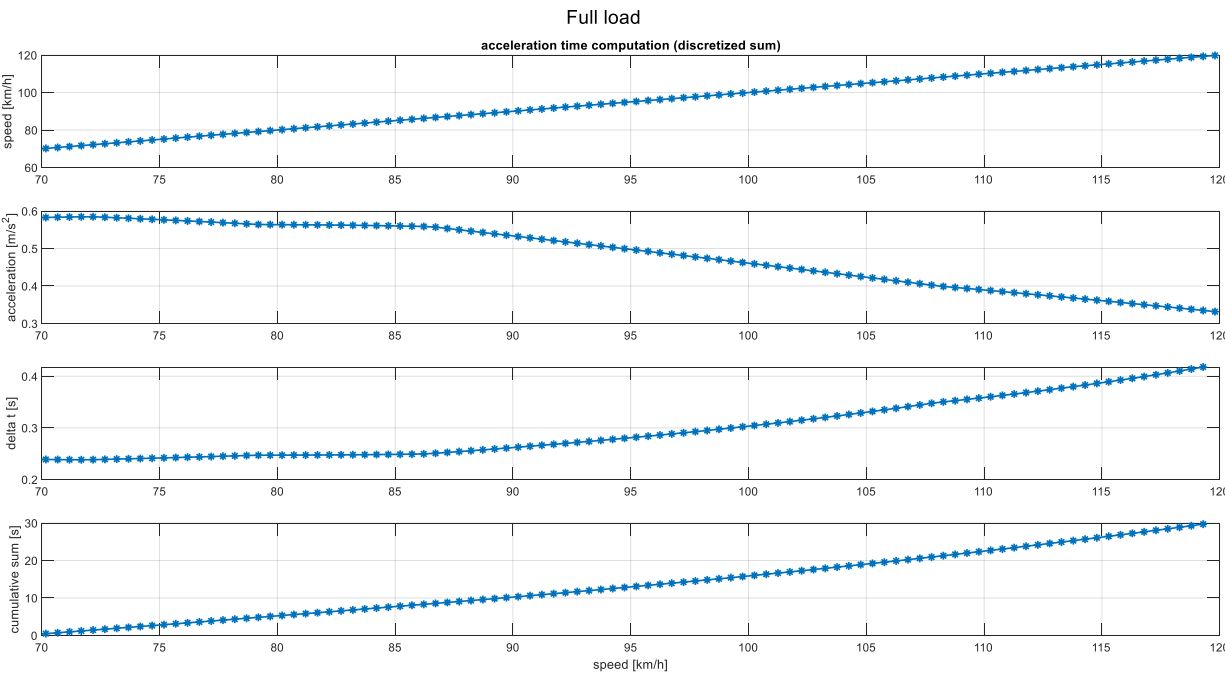


Figure 14

It is possible to observe for both cases that the acceleration decreases with higher speeds, the extra power given from the engine is reduced. The acceleration clearly varies in a non-linear way, highlighting the nonlinearities of the longitudinal dynamics as considered.

Acceleration time:

- Running order: 30.191 s
- Full load: 42.315 s

3 Conclusion

In this lab, the request was to analyse the general performance of a motor vehicle. Specifically, after having collected the data of the chosen vehicle (FIAT Panda of 2010), the analysis verted on determining the gear ratios in two load conditions as well as maximum speed and the time required to accelerate the vehicle.

The first request was, after having estimated the efficiencies of the different transmission components, to evaluate the resistant power considering the resistant forces due to aerodynamics, rolling and the slope of the road.

Afterwards, the highest and lowest gear ratios were evaluated. Specifically, for the first case, the ratio was varied as function of the speed until the intersection of the resistant and traction curves was located at the maximum point. The second one was instead evaluated in order for the car to climb a road with slope of 45%. This was done imposing the tangency between the traction and resistant power curves.

The intermediate ratios were then evaluated and compared with the ones provided by the manufacturer.

Finally, the top speed for the 2 conditions and the time needed to accelerate the car from 70 to 120 km/h was evaluated. It can be noticed that as the speed increases, the acceleration decreases. Naturally, due to higher resistant forces, the top speed for the condition at full load is lower and the required time to accelerate the car is instead higher.